2.1 – Bohr Models and Lewis Dot Structures

1) How did the Bohr model improve upon Rutherford's nuclear model?

2) Draw the Bohr model for each element listed below.

a) Sodium b) Carbon c) Argon

d) Aluminum e) Oxygen f) Potassium

3) How do you determine the number of valence electrons for an element?

4)Draw the Lewis dot structures for the following elements

a) Iodine b) Magnesium c) Fluorine d) Sodium e) Francium

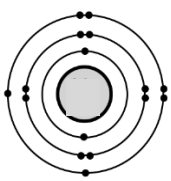
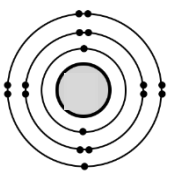
5) Identify the element

a) This element has 3 valence electrons and is in period 2. _____

b) This element has 7 valence electrons, and its most abundant isotope contains 10 neutrons. ____

c) This element has 8 valence electrons and is in period 3. ____

d) What elements are represented below by the Bohr models?

 _____  _____

Topic 2.2 - Quantum Mechanical Model and Electron Configuration

1. Write the electron configuration for Be, Mg, Ca. What is the similarity in the electron configuration of the outermost electrons in these elements?
   Be:

   Mg:

   Ca:

   Similarity:

2. Write the electron configuration for the following: (Circle the valence electrons)
   N:

   Cl:

   Mn:

   Cd:

3. Write shorthand electron configuration for the following:
   Sr:

   As:

   S:

   Al:

4. Identify the following elements from the electron configuration. (Circle the valence electrons)
   $1s^2 2s^2 2p^6 3s^1$:

   $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^3$

5. Are the following electron configurations correct? If not, correct.
   $1s^2 2s^2 2p^6 3s^1 3p^6 4s^2 4d^{10} 4p^6 5s^1$

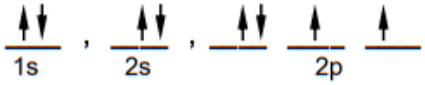
   $1s^2 2s^2 2p^6 3s^3$

   $[Ar] 5s^2 4d^{10} 5p^5$

2.3 - Electron Configuration and Orbital Diagrams Continued

There are three ways to indicate the arrangement of electrons around an atom:

1. Orbital Filling Diagram (gives the most information)

   Ex. O_2 

2. Electron Configuration (quicker to draw than orbital filling diagrams)

   Ex. O_2 $1s^2\ 2s^2\ 2p^4$

3. Electron Dot shows only the valence *(outer energy level)* electrons

   Ex. Oxygen atom 

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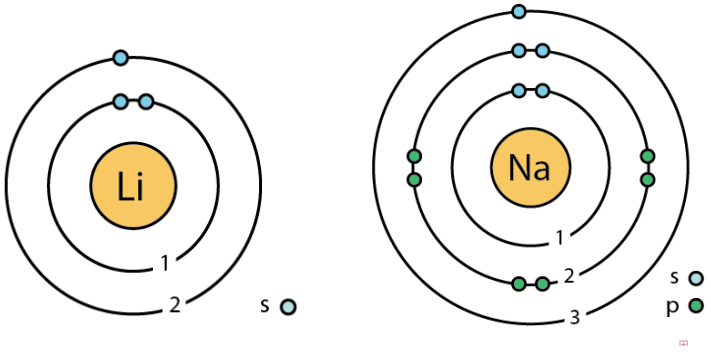
1. Write orbital filling diagrams, electron configurations, and electron dot diagrams for the following elements.

Table:

Element	Orbital Filling Diagram	Electron Configuration	Electron Dot Diagram
a. Boron			
b. Silicon			
c. Sulfur			
d. Calcium			
e. Iodine			
f. Rubidium			
g. Chromium			
h. Gallium			

Topic 2.4 - Periodic Trends: Ionization Energy and Atomic Radius

1) The following Bohr models indicate atoms of Lithium and Sodium.



a) On both models, circle the electron(s) that would be removed if the atoms were ionized.

b) What charge would both atoms form when they become ions? _____

c) Determine which atom would require more energy to remove those electrons. Use principles of atomic structure and coulomb's law in your explanation.

2) The ionization energy trend is that it takes more energy to remove an electron as you go from the left to the right (across a period.) Explain why the ionization energy increases using principles of atomic structure and coulomb's law.

3) The ionization energy trend is that it takes more energy to remove an electron as you go up a group on the periodic table. Explain why the ionization energy increases using principles of atomic structure and coulomb's law.

4) The atomic radius trend is that atoms get smaller as you go from the left to the right (across a period.) Explain why the atomic radius decreases using principles of atomic structure and coulomb's law.

5) The atomic radius trend is that atoms get smaller as you go up a group on the periodic table.Explain why the atomic radius decreases using principles of atomic structure and coulomb's law.

6) Rank the elements below in order of increasing ionization energy.

List 1: B N O Li Be F

List 2: N As Bi P Sb Mc

7) What is a larger atom?

a) K or Ca b) P or N c) Al or Ar d) F or Br

Topic 2.5 - Periodic Trends: Electronegativity and Successive Ionization Energy

Word bank: decreases, increases, successive, ionization, electronegativity

1. As you move across a period from left to right, the electronegativity of elements generally _____.
2. The _____is a measure of an atom's ability to attract electrons.
3. Successive _____energies increase with the removal of each electron.
4. As you move down a group, the electronegativity of elements generally _____.
5. _____ionization energy refers to the amount of energy required to remove electrons one by one from an atom.

6. Rank the following elements by increasing atomic radius; carbon, aluminum, potassium and oxygen.

7. Rank the following elements by increasing electronegativity; sulfur, oxygen, neon and aluminum.

8. Why does fluorine have a higher ionization energy than iodine?

9. Why do elements in the same family generally have similar properties?

10. Indicate whether the following properties increase or decrease from left to right across the periodic table.
    a. Atomic radius

    b. First ionization energy

    c. electronegativity